

BUILDING & COMMISSIONING A REMOTE WTP

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Owner Reference Resource

Purpose

This resource is useful enough to help asset owners identify issues, keep useful records, and decide when to escalate – it is not a repair guide, design guide, or compliance certificate.

A plain-language reference on what councils, government program managers, and asset owners should expect from the construction, commissioning, and handover of a water treatment plant in remote and regional Australia.

Important Scope Note

This guide is general information only. It is not a construction methodology, procurement model recommendation, contract advice, or regulatory interpretation.

Use it to frame what to expect from a WTP delivery contractor and what should arrive in your handover pack. Do not use it to specify construction methods, set acceptance criteria, or interpret defects-liability obligations.

What This Guide Helps You Do

- Understand the construction-to-operations journey for a modern WTP.
- Recognise what good commissioning looks like before responsibility transfers to your operations team.
- Know what should be in a complete handover pack.
- Identify warning signs that a project is heading for a poor handover.

What This Guide Does Not Cover

- Construction methodology selection.
- Procurement model selection (D&C, ECI, traditional).
- Contract terms or defects-liability negotiation.
- Regulatory compliance interpretation for a specific scheme.

Why Remote WTP Projects Are Different

A WTP delivered in a metropolitan or near-metropolitan setting and a WTP delivered in a remote or regional setting are not the same project at different scales. Remote delivery introduces structural differences that shape every other decision:

- **Distance from suppliers** – long lead times for chemicals, consumables, and replacement parts.
- **Single freight windows** – wet-season road or barge closures can cut the site for weeks.
- **Weather closures** – cyclones, monsoonal rainfall, and flooding compress the buildable calendar.
- **Limited local trade base** – most specialised trades fly in; sequencing and accommodation are part of the programme.
- **Logistics and accommodation** – workforce camp, mess, freight staging, and crane access shape the construction sequence.
- **Community engagement** – local engagement, cultural protocols, and Indigenous community considerations sit alongside the technical scope.
- **Single point of failure operations** – once handover happens, the local operator is often the only person on site. There is no backup shift.
- **Specialist competency gap** – process tuning, SCADA configuration, and PLC fault diagnosis usually require a fly-in specialist.

BUILDING & COMMISSIONING A REMOTE WTP

The owner's job is to make sure these realities are present in the design narrative, the construction programme, and the handover plan — not absorbed by the operator after handover.

From Construction to Operations: The Project Lifecycle

A modern WTP project moves through a defined lifecycle. Each stage has its own success criteria; failures upstream are most expensive when they surface at cutover.

1. Design Confirmation and Procurement Lock-In

Early in the construction contract, design is confirmed against the source water characterisation, equipment is specified, and long-lead procurement is locked in. Validation packages — for UV, for chemical systems, for SCADA — should be defined at this stage, not at commissioning.

2. Construction and Trade Integration

Civil, structural, mechanical, and electrical trades are sequenced around long-lead deliveries. Hold Points and Inspection and Test Plans (ITPs) document the integration of vendor packages with site works. Concrete tanks, equipment skids, electrical room and chemical bunds typically share interface points that need to be programmed, not improvised.

3. Dry Commissioning

Before any water moves through the plant, every input/output point is verified, every control loop is exercised, every alarm is tested, and every interlock is proven. The "Plant Healthy" signal philosophy — the master interlock that prevents treated water leaving the plant when any barrier is unavailable — is set up and tested here.

4. Wet Commissioning

Water moves through the plant in stages. Chemical dosing is proved. Filter performance is verified. UV is validated against its protocol. Chlorine residual is established. Ct is demonstrated at peak flow. This stage often surfaces the issues that the dry commissioning could not — and those issues are easier to fix while the contractor is still on site.

5. Proof of Performance

A defined sustained-run period — days or weeks — confirms that the plant meets ADWG performance under realistic operating conditions, not just at peak conditions. Microbial log reduction is demonstrated. Treated water quality is recorded against design.

6. Cutover

The point at which responsibility transfers from the contractor to the operator. The cutover criteria should be agreed in writing in advance. The operator should already have hands-on time on the actual plant before cutover — not after.

7. Decommissioning of Legacy Assets

Where the new plant replaces an existing plant, the legacy plant is isolated, redundant systems are removed, and as-constructed records are updated. This is project work, not operations work.

What "Good Commissioning" Looks Like

- Independent validation of disinfection equipment witnessed and documented.
- Test results documented, not asserted.
- Cutover criteria agreed before commissioning starts — not negotiated under pressure at the end.
- Operations team trained on the actual plant, by the contractor, before cutover.
- Snag list closed before cutover. Items that cannot be closed are documented as defects with owners and dates, not deferred without record.

BUILDING & COMMISSIONING A REMOTE WTP

The Handover Package – What an Owner Should Receive

A complete handover package gives the operations team everything they need to run the plant from day one and an audit trail for the next regulator visit. At minimum:

Item	Why it matters
As-constructed drawings and P&IDs	The operator needs the actual plant, not the design plant.
O&M manuals	Vendor documentation collated, indexed, and accessible to the operator.
Validation evidence	UV protocol, Ct calculation, ADWG performance run results.
SCADA documentation	Tag list, alarm philosophy, control narratives, alarm response procedures.
Training records	Documented competency confirmations for each operator.
Recommended spares list	With supplier, part number, and lead time – not a generic shopping list.
Chemical specifications	Active ingredient, concentration, supplier, lead time, minimum stock level.
Maintenance schedule	Daily, weekly, monthly, quarterly, annual tasks with responsible owner.
Emergency response procedures	Power failure, UV failure, gas leak, cyclone, SCADA failure, chemical depletion.
Defect lists and warranty register	Every outstanding item, every warranty period, every contact.
Cyclone and wet-season preparedness plan	Where applicable to the site.

If any of these is missing at handover, the operator inherits a plant they cannot defend at audit.

Operating a Remote WTP – Realities Owners Should Plan For

- **Chemical lead times and minimum stock levels** – extended storage windows are normal for remote sites; running out is not an option, and resupply during the wet season may be impossible.
- **UV warm-up bottleneck** – every restart includes a warm-up period during which no treated water can be produced. Reservoir levels need to be managed around this.
- **"Plant Healthy" interlock philosophy** – operators must never bypass it. Bypassed interlocks are how plants quietly fail their compliance position.
- **Cyclone and wet-season preparedness** – generator load testing, fuel reserves, chemical stocks, secured equipment, maximum reservoir levels, emergency contacts confirmed before each season.
- **Critical spares vs ordered-on-failure spares** – some items must be on the shelf; some can wait. The list should be explicit, not implicit.
- **When to use the local operator vs when to escalate** – routine operation belongs locally. PLC faults, SCADA configuration, process optimisation, and HAZOP-level issues need a specialist.

Warning Signs of a Poor Handover

- No validation evidence – UV, Ct, or ADWG performance run records cannot be produced.
- Missing or out-of-date as-constructed drawings.
- Training delivered as PDFs rather than on the actual plant.
- Snag list "to be closed during operations" – the operator inherits the contractor's unfinished work.
- Spares list deferred to a later project phase.
- No agreed cutover criteria, or criteria changed at cutover under time pressure.
- No documented operator-level emergency procedures.
- No clear post-handover support pathway – who do we call when something goes wrong?

Seven Rules of Remote WTP Operation

Carry these into every operations meeting:

1. **Plant Healthy is everything.** Plant Healthy ON = water can be produced. OFF = stop. Never bypass it.
2. **UV is the protozoa barrier.** No treated water leaves without UV In-Service. The warm-up period is non-negotiable.

BUILDING & COMMISSIONING A REMOTE WTP

3. **Respect the seasons.** Wet season is high demand on chemicals, filters, and sludge handling. Prepare before it starts.
4. **Chemical supply is your lifeline.** If chemicals run out, the plant stops. Order early, hold buffer stock, plan around freight.
5. **Chlorine gas demands respect.** AS2927 is not optional. Gas leak detection, ventilation, breathing apparatus, drilled procedures.
6. **Alarms mean something.** Every alarm requires investigation. Repeat alarms are a fault, not a habit.
7. **Know when to call for help.** Routine operations belong locally. PLC faults, SCADA configuration, and process optimisation need specialists.

Owner Decision Prompts

Questions to ask the contractor, consultant, or program manager before contract signature, commissioning, and cutover:

- What are the cutover criteria, and who agrees them?
- When does our operations team get hands-on time on the actual plant?
- What validation evidence will we receive at handover, and in what form?
- What chemical and consumable stock levels are recommended for our location, and on what assumptions about resupply?
- Who do we call when something goes wrong post-handover, and what is the response time?
- What is the snag-list close-out plan, and what is the consequence of items not closed by cutover?

Next Step

Remote WTP delivery is the discipline of treating handover as the start of the operating life — not the end of the project. Most projects that struggle in operations did not budget enough time, evidence, or training into the cutover.

PC Water Solution plans, builds, commissions, and supports water treatment plants in some of Australia's and the Pacific's most remote locations. We can engage at any point — design or tender review, construction support, commissioning oversight, handover-package review, or end-to-end project delivery — and we treat the operator who inherits the plant as the audience that matters most.

Plan a remote WTP project with PC Water Solution — call 1300 029 804, email info@pcwater.com.au, or visit pcwater.com.au.